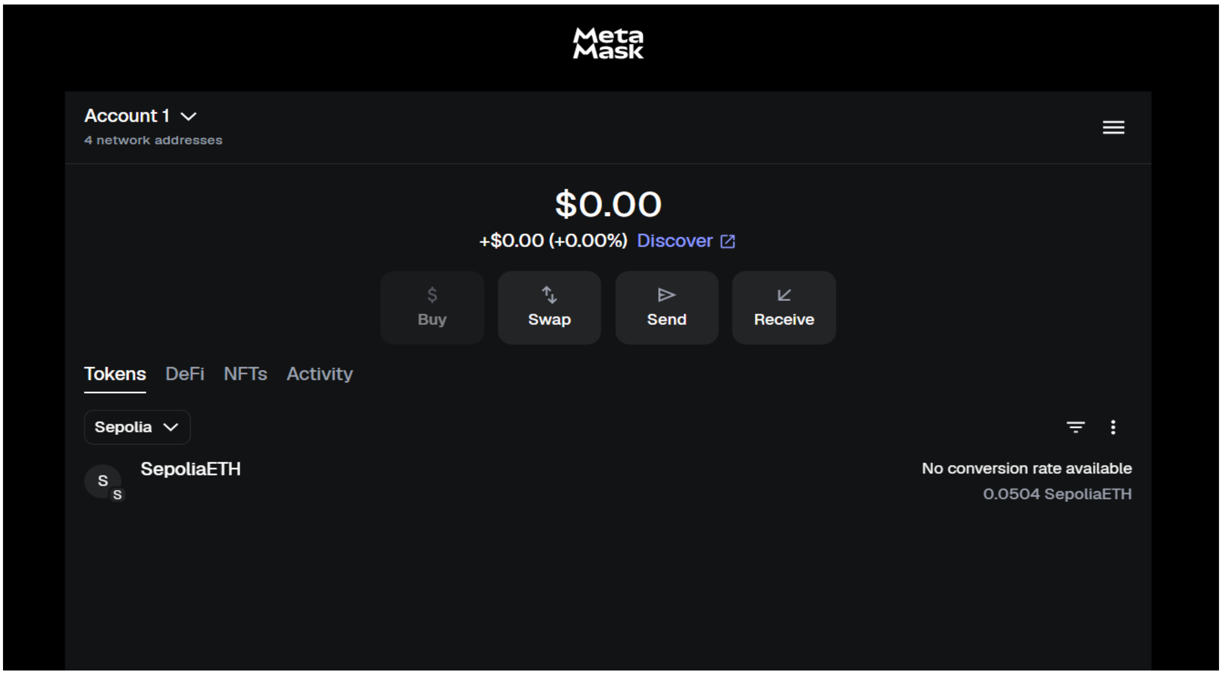
Web3 Basics Submission Document

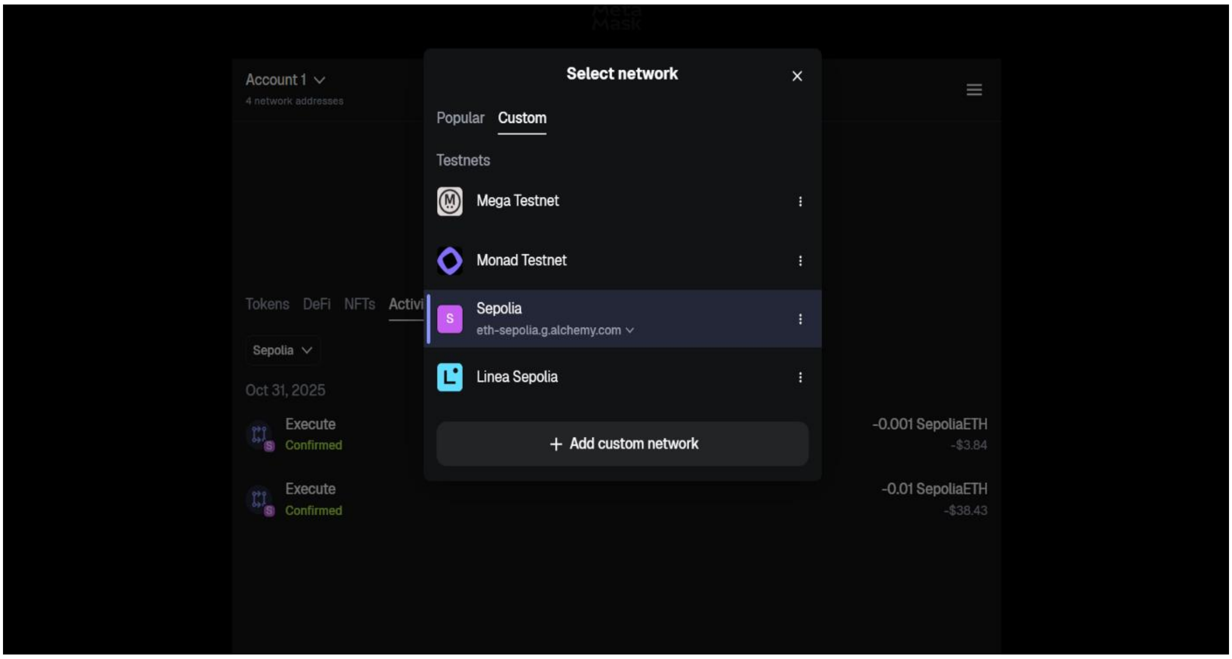
1. Documentation

1.1 Screenshots:

- Metamask Setup



- Testnet Configuration



- Faucet Transaction

Sepolia PoW Faucet

The faucet is currently experiencing unusually high mining activity (>333 kH/s). For higher rewards, I recommend visiting again when the activity is lower.

Claim Rewards

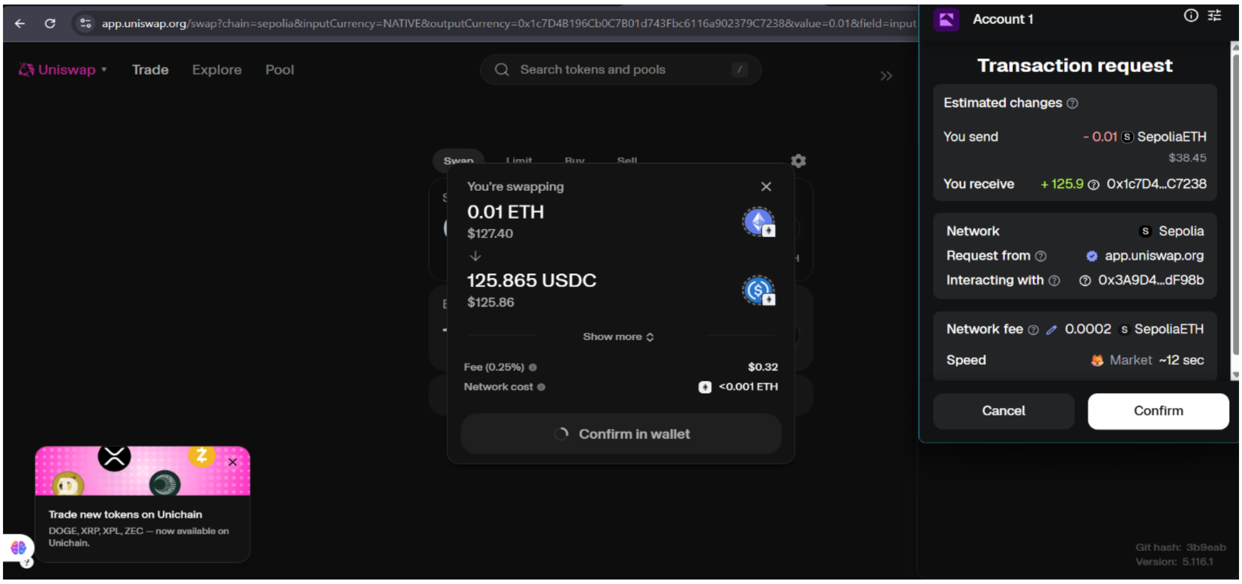
Wallet:	0xe85303b528b5314611585ad5d1335c6b4dd89c66
Amount:	0.05 SepETH
Timeout:	-

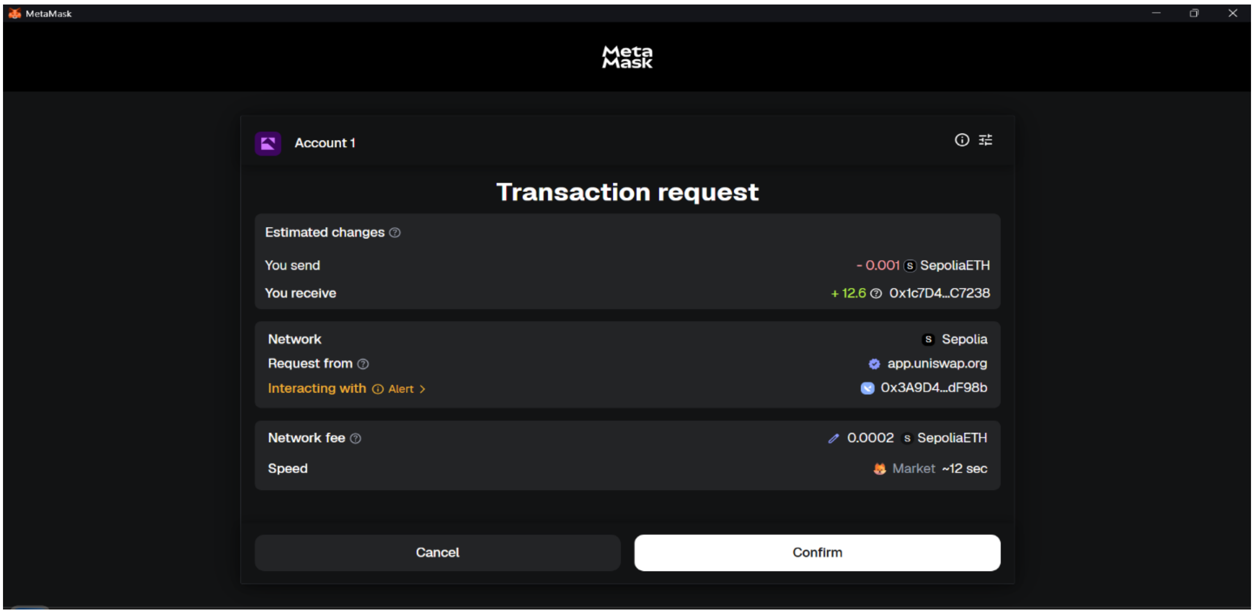
Claim Transaction has been confirmed in block #9516866!
TX: 0xe7eea995cb0bfcbd7f7f1151ef338698398588de02dc6f11e0d4dede2f7feca0

Did you like the faucet? Give that project a Star 5,241
Or support this faucet by sharing your result with a Tweet Post

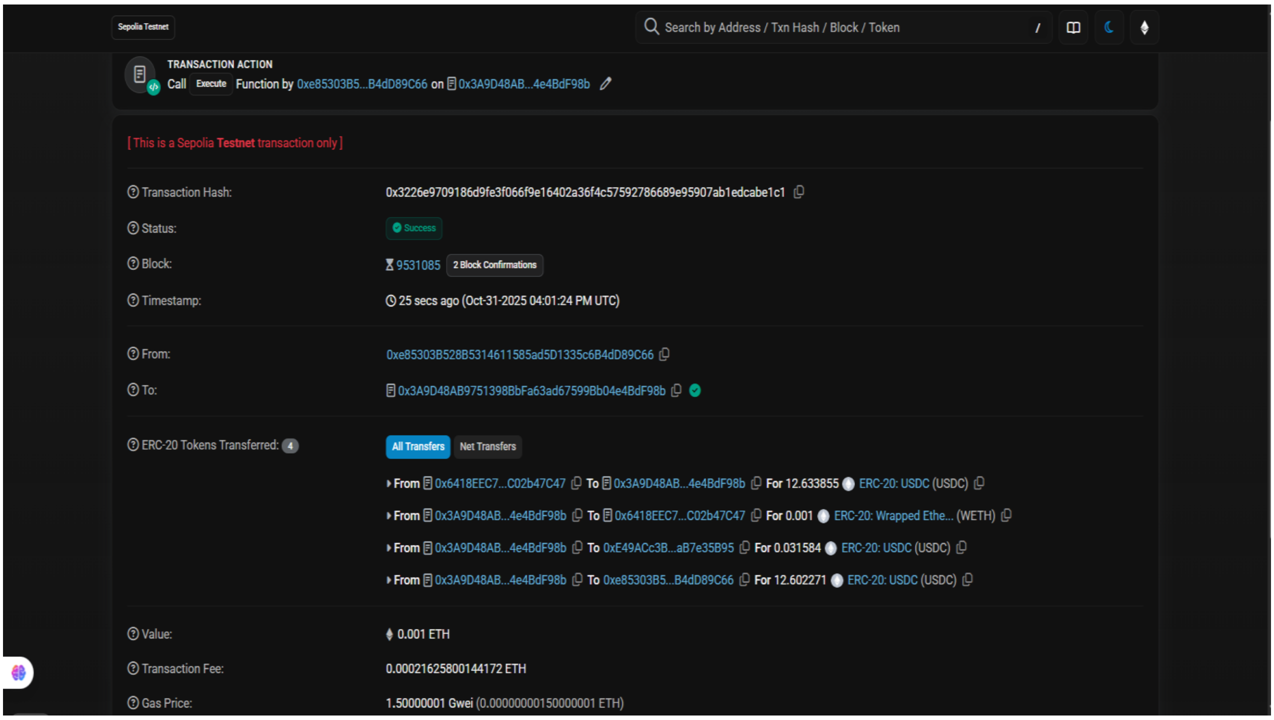
Return to startpage

- DApp connection





- Completed transaction



1.2 Wallet Public Address:

- **0xe85303B528B5314611585ad5D1335c6B4dD89C66**

1.3 Transaction Hash(es):

- **0x3226e9709186d9fe3f066f9e16402a36f4c57592786689e95907ab1edcabe1c1**

1.4 Etherscan Testnet Explorer Link(s):

- https://sepolia.etherscan.io/address/0xe85303B528B5314611585ad5D1335c6B4dD89C66

2. Written Reflection

During this Web3 Basics exercise, I gained hands-on experience with blockchain technology and decentralized applications (DApps). I learned how blockchain operates as a distributed ledger where data is immutable and publicly verifiable. Key concepts like transactions, blocks, miners, and consensus mechanisms helped me understand how trust is achieved without central authority. The difference between centralized and decentralized applications became clear: centralized apps depend on a single server or authority, while DApps run on blockchain networks, offering transparency, security, and censorship resistance. Decentralized systems give users ownership of data and assets, unlike traditional apps controlled by companies. Smart contracts play a vital role in DApps as self-executing programs stored on the blockchain. They automate processes without intermediaries, ensuring transparency and reliability. For example, transactions in my DApp were governed by smart contracts that executed based on pre-defined logic. Security is critical when handling crypto wallets. Users must safeguard their private keys and seed phrases since losing them means losing wallet access permanently. I also learned to verify URLs, use trusted networks, and avoid sharing sensitive information. One of the main challenges was configuring

MetaMask with the Sepolia test network and obtaining test ETH. I overcame this by using faucet sites and carefully checking RPC settings. Another issue was transaction delays, which I resolved by increasing gas limits. This project deepened my understanding of blockchain ecosystems, DApp structure, and secure crypto interactions.

3. Technical Summary

- Testnet Used Sepolia Testnet.
- DApp Interacted With Simple Ethereum Transaction Dapp.
- Types of Transactions Wallet connection and token transfer.
- Errors Encountered RPC configuration error, transaction delay.
- Troubleshooting Steps Verified network RPC settings, increased gas limit, reconnected wallet.